2.Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int src, dest, bytes;
    char buffer[1024];

    src = open("input.txt", O_RDONLY);
    if (src < 0) {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while ((bytes = read(src, buffer, sizeof(buffer))) > 0) {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);
    printf("File copied successfully.\n");
    return 0;
}
```

knsat@SRITHIKA: ~

```
knsat@SRITHIKA:~$ echo "Hello, this is a test file for the OS lab assignment." > input.txt
knsat@SRITHIKA:~$ nano copy.c
knsat@SRITHIKA:~$ gcc copy.c -o copy
knsat@SRITHIKA:~$ ./copy
File copied successfully.
knsat@SRITHIKA:~$ cat output.txt
Hello, this is a test file for the OS lab assignment.
knsat@SRITHIKA:~$ echo "This is sample text for strace." > sample.txt
knsat@SRITHIKA:~$ strace cat sample.txt
Command 'strace' not found, but can be installed with:
sudo apt install strace
knsat@SRITHIKA:~$ sudo apt install strace
[sudo: authenticate] Password:
Installing:
  strace

Installing dependencies:
  libunwind8

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 62
  Download size: 737 kB
  Space needed: 2660 kB / 1024 GB available

Continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [678 kB]
Fetched 737 kB in 0s (1659 kB/s)
Selecting previously unselected package libunwind8:amd64.
(Reading database ... 40672 files and directories currently installed.)
Preparing to unpack .../libunwind8_1.8.3-0ubuntu1_amd64.deb ...
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...
Selecting previously unselected package strace.
Preparing to unpack .../strace_6.19+ds-0ubuntu5_amd64.deb ...
Unpacking strace (6.19+ds-0ubuntu5) ...
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Processing triggers for man-db (2.13.1-1build1) ...
```